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PREDICT NEW YORK CITY AIRBNB PRICES

Linear Regression Project | One-Page Summary Report

Objective

This project develops a Linear Regression model to estimate the nightly price of a New York City Airbnb listing using available listing information. The aim is to assess how well a simple, data-driven model can estimate prices for listings it has not seen during training.

Dataset and Methodology

The analysis uses the New York City Airbnb Open Data dataset specified in the project instructions. Exploratory data analysis examined the dataset's dimensions, columns, data types, and missing values. The **last_review** column was removed; missing values in **reviews_per_month** were investigated and handled; and identifier/descriptive columns were dropped. Outliers were inspected with boxplots and histograms, then listings with non-positive prices, prices above the 95th percentile, or minimum stays of 365 nights or more were excluded. The high-cardinality **neighbourhood** column was removed, while **neighbourhood_group** and **room_type** were one-hot encoded with the first category dropped. The data was split into 80% training and 20% testing data using **random_state=42**. A Linear Regression model was trained on the training set and evaluated on the held out test set.

Model Performance

Enter the actual metric values printed by your Google Collab notebook in the boxes below before submission. These values cannot be determined from the project instructions alone.

R-SQUARED (R^2)	MEAN ABSOLUTE ERROR (MAE)
38.14059542509071	**$38.14059542509071**

What the Metrics Mean

R^2: This score describes how much of the variation in listing prices the model explains compared with a baseline that predicts the average test-set price. A score closer to 1 generally indicates a stronger fit; a negative score means the model performs worse than that baseline on the test data.

MAE: This is the average absolute difference between the predicted and actual nightly prices. For example, an MAE of $20 would mean predictions are off by about $20 per listing on average in absolute dollar terms. Replace this example with your actual MAE when interpreting the model.

Residual Analysis and Findings

The actual-versus-predicted plot should be reviewed to see how closely predictions follow the perfect-prediction diagonal. The residual plot shows whether errors are scattered around zero or follow a pattern. The cheaper-versus-more-expensive listing comparison should be completed using the MAE and percentage-error outputs from Step 7. Record which group has the lower error and briefly explain the observed pattern here.

Conclusion

This project demonstrates a complete basic machine-learning workflow: data auditing, cleaning, outlier treatment, categorical encoding, train-test splitting, model training, evaluation, and residual analysis. The model's practical usefulness should be judged from the actual test-set metrics and error plots. Its output is an estimate, not a guaranteed market price.